Supplementary Material: Simulated Drives over Measured Road Profiles for Car-Based IRI Estimation

S1. Protocols, evidence classes and deviations

Three protocols. The first protocol (`evaluation/FINAL_EVALUATION_PROTOCOL.md`) and the model configurations (`config/models.json`) were frozen before any test evaluation; they define the estimators, the simulator V0, the leave-one-route-out split and the metrics (evidence class C1). After an independent review of the first version of this paper, a second protocol (`evaluation/PROTOCOL_V2.md`) was frozen before any of the new analyses was run. It lists the simulator variants V1–V10 with their seeds, the calibration and equal-budget designs, the fixed-road speed analysis, the inference rule (a paired difference is "supported" only if its 5 km-block interval excludes zero), the diagnostics and the controlled vehicle-number experiment, and it forbids choosing any variant, block size, k, draw, seed or subset after seeing its result. Four addenda (E5–E8, Section 8 of that file) were written after some of these results were known and are reported in full: the equal-budget arms for V1–V8, the paired per-draw comparison with real-only training, the speed scaling of raw vibration in simulation, and paired elasticity differences. After a second independent review, a third protocol (`evaluation/PROTOCOL_V3.md`, dated 2026-10-10) was frozen before its analyses were run. It restores the joint-training arm of the first protocol ("simulation-pretrained + real", implemented for LightGBM with equal total weight for simulated and real units, as in our first analysis round, which reported it for one simulation set and five draws; Table S20) for V0–V8 and V10 on the 20 shared draws (class R), and adds the test-route intervals of the calibration comparison, the band-limited retraining, all band associations, the level decomposition by road class, a paired test of within-route ranking, robustness rows for the speed elasticities, a no-simulator index, the pass density of the paired data and relative errors (class E). The reviewer had computed several of these on the same data before the protocol was written (the joint-training arm for V0, V1, V3 and V5, the band-limited retraining, the paired ranking test, the speed robustness rows and the no-simulator index), so none of them is confirmatory; where they overlap, the reviewer's values were reproduced (joint training identically for every fold and draw). Its hypotheses were stated before the runs: that joint training beats real-only training for every variant up to 10 km and stays within 0.02 m/km of it at 20 km (holds for V0–V8 and holds with V10), and that retraining below 10 Hz halves the spread of the sensor variants' levels while dropping the 10–15 Hz band alone changes it by less than 20 % (holds) and leaves V4 outside their range (holds).

Evidence classes. C1: fixed before any test result. R: pre-specified re-analysis of routes whose results were already known. E: exploratory diagnostic. S: new controlled simulation with freshly generated test data. All real-data results of the main paper except the comparison of V0 with the real-trained estimators are R or E.

Deviations in the third analysis round.

- The no-simulator comparison also reports the three-parameter index fitted to V0 simulated units with a level offset, which reproduces a second-round result (Table S15); it is not used for any claim.
- The band associations were also computed against the motorway-minus-urban level difference; this target was not in the protocol, is kept in the result file and is not used for any claim.

Deviations in the second analysis round.

- V7 and V8 enter the sensor factorial with a random half of their passes (seed 7) to match V1 and V2 in size; decided before running. Results for the full sets are in Table S14.
- The pre-specified aliasing check within one telematics unit (TMU#10) was infeasible: that unit logged at 52.6 Hz only on M13 and at 333 Hz only on CPH1 and M3. TMU#12 on M3 had no common segment × speed cell either; the check was replaced by an all-unit comparison on common cells, which is confounded with device and mounting.
- The keys (segment, task, pass) of the real units are not unique (12 duplicates where a ring-road pass covers a segment twice); earlier predictions are therefore aligned by position and checked against the labels.
- Units shorter than one Welch window have no spectrum and are excluded from the spectral cell means.

Deviations in the first analysis round.

- *Robustness experiment.* The first version drew new random passes for every perturbation level; it was re-run with common random numbers (variance reduction). The first version is kept in `results/sim/robustness_v1_independent_draws.csv`.
- *Compute allocation.* Zero-shot CNNs (three seeds, two simulation sets) and few-shot CNNs were limited before their results were known.
- *Seeds.* Neural-net metrics are those of the 5-seed ensemble prediction in the main paper; per-seed means are in `results/v2/summary_real_estimators.csv`.

- *Sensor statistics.* The telematics configuration of the simulator (55–62 Hz, 0.01 g, tilt s.d. 6°) was set from a preliminary inspection of the unlabelled LiRA-CD files. A complete re-measurement (`evaluation/lira_sensor_stats.py`) gives nominal rates of 52.6 Hz (66 % of passes) or 333 Hz, 0.01 g quantisation in 81 % of passes and tilts with median 10° (maximum 41°). V0 was not changed; the matched configurations are the variants V5 and V6.

Statements of the first version that were wrong and are corrected. (i) Real-only training was said to need "about 20 km" to match simulation; the comparator was a different simulation set, and the profile budgets were unequal (replaced by the equal-budget design). (ii) A caption called the analysis "pre-registered"; it was frozen in advance by us, not registered. (iii) "Sampling rate matters most" among sensor perturbations; the holder resonance moves the bias more. (iv) The CNN deficit was presented as general; it sits on one route. (v) Transfer was attributed to the breadth of the sensor model; the sensor factorial contradicts this (main paper, Section V-B). (vi) Urban errors were said to be smaller than the disagreement between two profilometers; that disagreement is driven by a heavy tail (Table S13), and the comparison was removed.

Statements of the second version that were wrong or incomplete and are corrected. (i) V0 was said to match the level "only because" half of its passes carried a phone-holder resonance; with the same sensor mixture and one vehicle (V4) the bias is +0.40 m/km, so the vehicle population is a second ingredient, and V0's matched pooled level hides a motorway level of +0.145 (Section V-B). (ii) The bias was said to follow a deficit of simulated vibration at 10–15 Hz; it also correlates strongly with the 5–10 Hz and overall-RMS ratios (Table S23), and only the removal of every feature with content above 10 Hz reduces the spread of the sensor variants (Table S24). (iii) With 10 km or more of paired drives, real drives alone were said to beat the three variants with the largest deficit; this holds for a level offset, not for joint training (Section V-D). (iv) The untrained statistic was said to rank segments within routes as well as the trained estimators; paired per route, all differences are within 0.05, but real-trained LightGBM ranks urban segments better and motorway segments slightly worse (Table S5). (v) The abstract mixed within-route and pooled MAEs, and "k km of paired drives" was not defined with its pass density. (vi) A segment-grid bug restricted M7 to 119 of the 228 M13 segments (MAE 0.169, now 0.168 m/km); the other M7 folds change slightly because their transmissibility is fitted on more M13 segments (Table S12). (vii) The anchor to Bajic et al. claimed a "different profiler"; what differs is the reference survey and its processing (Table S10). (viii) The mounting of the telematics unit was called unknown; the dataset article describes it (Section III). The P79 survey dates of CPH6 and the ARAN survey dates were misstated (Table S13).

S2. The earlier pipeline

Labels. On the 7 LiRA-CD route-directions whose profile and IRI files are consistent, our IRI implementation reproduces the IRI reported by the P79 (an algorithm check: both are computed from the same laser profiles) with $r = 1.000$, bias +0.018 m/km and MAE 0.019 m/km over 1,520 segments of 100 m (Fig. S1a). For the eighth file (CPH6, direction HH) the reported IRI cannot be aligned with the file's own profile (best $r = 0.13$ over shifts of ± 3 km, with or without reversal, against $r = 1.000$ at zero shift for two control files); all labels are therefore recomputed from the profiles. The IRI routine of the earlier pipeline is also correct (bias −0.002 m/km), but its curvature-adaptive mesh correction is not: applied to the same real profiles, which contain no mesh facets, it lowers the IRI of urban segments by 28 % on average (bias −1.10 m/km) and that of motorway segments by 6 % (Fig. S1b). On the BeamNG corpus the correction removes 64 % of the summed IRI (mean 12.50 vs 4.51 m/km) and changes the condition class of 34 % of the segments; the earlier labels were a smoothed roughness index, and the uncorrected ray-cast profiles give implausibly high values, so the corpus has no valid IRI label.

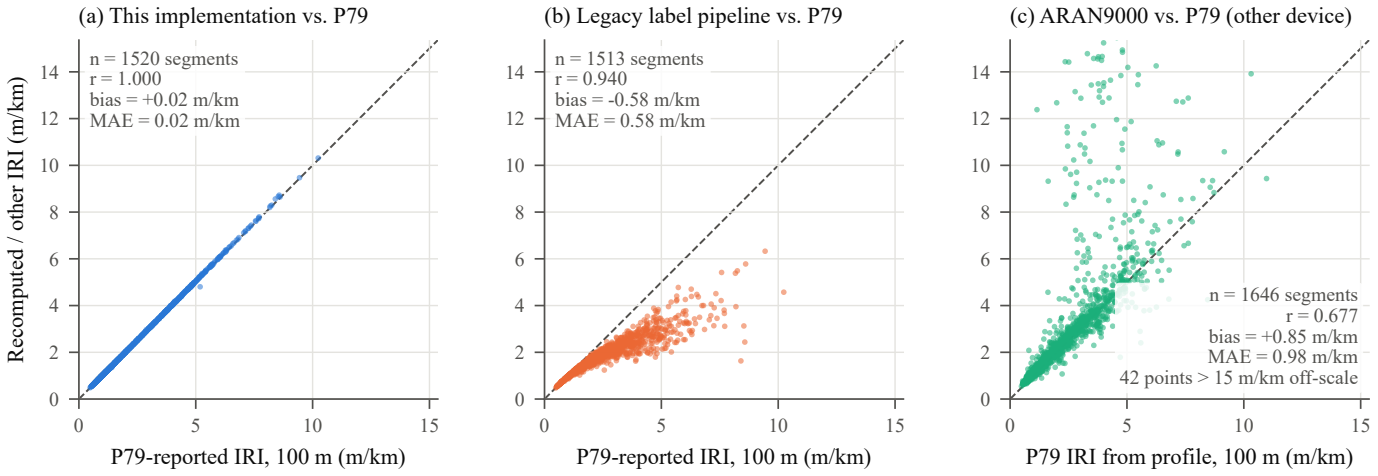


Fig. S1. Label validity on real profiles (LiRA-CD, 100 m segments, mean of both wheel paths). (a) IRI recomputed from the P79 laser profiles with our implementation vs the IRI reported by the P79. (b) The earlier label pipeline, including the curvature-adaptive mesh correction, applied to the same profiles. (c) A second profilometer (ARAN9000, other survey date) vs the P79; off-scale points are counted in the panel and included in the statistics.

Protocol. The corpus contains 19 manually driven trips of 3 BeamNG.tech vehicles on 3 maps; IMU, position and elevation were polled by unsynchronised threads, which produced exact sample-and-hold duplicates in 48–68 % of consecutive IMU rows per trip. Re-run with non-overlapping 100 m test segments and the earlier labels, the earlier recipe reaches MAE 1.67 m/km and Spearman $\rho = 0.80$ with the earlier split, in which early stopping and isotonic calibration use one trip on the test track and two further trips on the same track form the test set; under leave-one-map-out it reaches 1.99 m/km and $\rho = 0.66$, no better than an MLP on the hand-crafted features (1.90 m/km; Fig. S2). Without the calibration fitted on the test map, the exponentiated network outputs diverge on a few segments of unseen maps (MAE 251 m/km). The earlier reporting rule hid the same failure on the earlier split: discarding test predictions above 40 m/km removed 1–2 segments per seed and turned uncalibrated MAEs of up to 1,555 m/km into at most 1.94 m/km. When vehicle and map are both unseen (raw-profile labels), no estimator is substantially better than predicting the training median (MAE 7.60; vibration index 6.88; LightGBM 7.25 m/km). The earlier study's asymmetric Huber loss reduces to a scalar before Keras applies sample weights, so its "two-dimensional inverse-frequency sampling" did not re-weight individual samples; our re-implementation reproduces this behaviour.

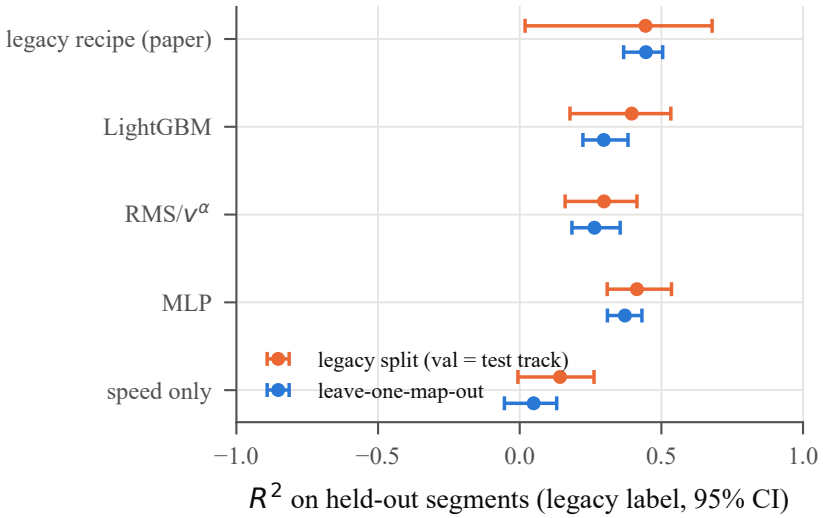


Fig. S2. BeamNG corpus, earlier label definition: R^2 on held-out segments under the earlier split (validation and test trips on the same track) and under leave-one-map-out (95 % cluster-bootstrap CIs).

Transfer to real cars. Trained on the BeamNG corpus and tested on LiRA-CD with the protocol of the main paper, LightGBM reaches MAE 1.36 m/km with the earlier labels and 1.97 m/km with raw-profile labels (within-route ρ 0.35 and 0.32; Table S14); its estimates rise steeply with speed on a fixed road (elasticity $+0.79$ and $+1.39$). The three-parameter vibration index trained on the same corpus ranks segments within routes far better (ρ 0.83 with the earlier labels) but misses the level (MAE 0.85 m/km, bias -0.73 m/km; Table S15), the same shape-versus-level pattern as the parametric simulator. The evaluated BeamNG pipeline did not transfer under this protocol; whether a game engine with measured road profiles and validated labels would transfer was not tested.

PVS. On the Passive Vehicular Sensors dataset (no IRI; subjective surface and quality labels), speed separates asphalt from the other surfaces with leave-one-route-out AUC 0.97 [0.94, 0.99] and "bad" from "regular" windows within cobblestone and dirt with 0.75, whereas the earlier simulation-trained model reaches 0.41 [0.33, 0.50] on the latter task (Table S2). The subjective labels are better predicted by how slowly the vehicle was driven than by how much it vibrated, so this dataset cannot validate an IRI estimator.

TABLE S1
BEAMNG CORPUS: ALL ESTIMATORS AND SPLITS (SEGMENT LEVEL, CLUSTER-BOOTSTRAP 95% CIS OVER 100 M LOCATION CELLS).

Split	Label	Model	MAE (m/km)	Spearman	R2
S3: earlier split (val. and test on test track)	legacy	M0 constant (training median)	2.68 [2.04, 3.39]	–	-0.07 [-0.19, -0.00]
S3: earlier split (val. and test on test track)	legacy	M1 speed-only (LightGBM on speed features)	2.20 [1.65, 2.85]	0.47 [0.32, 0.59]	0.14 [-0.01, 0.26]
S3: earlier split (val. and test on test track)	legacy	M2a RMS only (alpha=0)	2.25 [1.71, 2.85]	0.67 [0.55, 0.75]	0.11 [-0.25, 0.34]
S3: earlier split (val. and test on test track)	legacy	M2b RMS/v^0.5	2.10 [1.60, 2.65]	0.71 [0.60, 0.79]	0.23 [-0.04, 0.42]
S3: earlier split (val. and test on test track)	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.03 [1.56, 2.55]	0.72 [0.62, 0.79]	0.30 [0.16, 0.41]
S3: earlier split (val. and test on test track)	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.02 [1.55, 2.53]	0.68 [0.57, 0.76]	0.29 [0.20, 0.36]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (all features)	1.88 [1.45, 2.36]	0.74 [0.63, 0.81]	0.40 [0.18, 0.53]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (no speed features)	1.97 [1.50, 2.48]	0.72 [0.60, 0.80]	0.35 [0.20, 0.50]
S3: earlier split (val. and test on test track)	legacy	M4 MLP	1.79 [1.34, 2.27]	0.76 [0.67, 0.83]	0.41 [0.31, 0.54]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN	2.39 [1.47, 3.58]	0.75 [0.65, 0.82]	-3.28 [-14.77, 0.43]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN (no speed input)	3.20 [1.69, 5.72]	0.69 [0.59, 0.77]	-17.73 [-75.41, 0.35]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (calibrated on val)	1.67 [1.29, 2.11]	0.80 [0.72, 0.85]	0.44 [0.02, 0.68]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (uncalibrated)	330.85 [1.43, 1043.50]	0.80 [0.72, 0.85]	-5902105.16 [-24835354.32, 0.63]
S1: leave-one-map-out	legacy	M0 constant (training median)	3.01 [2.61, 3.48]	-0.37 [-0.48, -0.25]	-0.21 [-0.30, -0.16]
S1: leave-one-map-out	legacy	M1 speed-only (LightGBM on speed features)	2.60 [2.26, 3.00]	0.31 [0.22, 0.39]	0.05 [-0.05, 0.13]
S1: leave-one-map-out	legacy	M2a RMS only (alpha=0)	2.33 [1.98, 2.74]	0.56 [0.51, 0.61]	0.10 [0.02, 0.18]
S1: leave-one-map-out	legacy	M2b RMS/v^0.5	2.15 [1.83, 2.52]	0.66 [0.61, 0.70]	0.22 [0.14, 0.31]
S1: leave-one-map-out	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.10 [1.79, 2.46]	0.67 [0.61, 0.71]	0.26 [0.18, 0.35]
S1: leave-one-map-out	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.13 [1.83, 2.49]	0.64 [0.58, 0.69]	0.28 [0.20, 0.36]
S1: leave-one-map-out	legacy	M3 LightGBM (all features)	2.06 [1.77, 2.41]	0.68 [0.63, 0.72]	0.30 [0.22, 0.38]
S1: leave-one-map-out	legacy	M3 LightGBM (no speed features)	2.07 [1.77, 2.42]	0.67 [0.62, 0.71]	0.30 [0.23, 0.38]
S1: leave-one-map-out	legacy	M4 MLP	1.90 [1.64, 2.22]	0.73 [0.69, 0.77]	0.37 [0.31, 0.43]
S1: leave-one-map-out	legacy	M5 1D-CNN	2.23 [1.79, 2.77]	0.70 [0.66, 0.74]	-3.33 [-13.32, 0.35]
S1: leave-one-map-out	legacy	M5 1D-CNN (no speed input)	2.64 [1.98, 3.60]	0.62 [0.56, 0.67]	-15.92 [-60.71, 0.22]
S1: leave-one-map-out	legacy	Legacy recipe (calibrated on val)	1.99 [1.76, 2.25]	0.66 [0.59, 0.72]	0.45 [0.37, 0.50]
S1: leave-one-map-out	legacy	Legacy recipe (uncalibrated)	251.39 [1.82, 773.08]	0.68 [0.62, 0.73]	-17368673.46 [-65761393.61, 0.37]
S1: leave-one-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.25 [-0.36, -0.14]	-0.15 [-0.23, -0.09]
S1: leave-one-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.49 [6.64, 8.46]	0.21 [0.12, 0.30]	-0.10 [-0.23, 0.00]
S1: leave-one-map-out	raw profile	M2a RMS only (alpha=0)	6.68 [5.90, 7.58]	0.49 [0.42, 0.55]	0.08 [-0.03, 0.17]
S1: leave-one-map-out	raw profile	M2b RMS/v^0.5	6.36 [5.62, 7.23]	0.54 [0.47, 0.60]	0.16 [0.05, 0.25]
S1: leave-one-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.34 [5.60, 7.20]	0.54 [0.46, 0.60]	0.17 [0.06, 0.25]
S1: leave-one-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.50 [5.72, 7.40]	0.48 [0.38, 0.56]	0.13 [0.01, 0.22]
S1: leave-one-map-out	raw profile	M3 LightGBM (all features)	6.02 [5.37, 6.73]	0.57 [0.50, 0.63]	0.26 [0.14, 0.35]
S1: leave-one-map-out	raw profile	M3 LightGBM (no speed features)	6.01 [5.38, 6.75]	0.58 [0.51, 0.63]	0.26 [0.15, 0.35]
S1: leave-one-map-out	raw profile	M4 MLP	5.70 [5.15, 6.30]	0.59 [0.52, 0.64]	0.34 [0.19, 0.44]
S1: leave-one-map-out	raw profile	M5 1D-CNN	8.77 [5.82, 14.01]	0.51 [0.44, 0.58]	-110.29 [-387.20, 0.22]
S1: leave-one-map-out	raw profile	M5 1D-CNN (no speed input)	9.75 [5.90, 16.77]	0.52 [0.44, 0.59]	-214.49 [-752.26, 0.13]
S2: leave-one-vehicle-and-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.23 [-0.34, -0.12]	-0.15 [-0.23, -0.09]
S2: leave-one-vehicle-and-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.61 [6.80, 8.55]	0.21 [0.11, 0.29]	-0.11 [-0.25, -0.00]
S2: leave-one-vehicle-and-map-out	raw profile	M2a RMS only (alpha=0)	7.27 [6.54, 8.09]	0.42 [0.36, 0.47]	-0.02 [-0.15, 0.07]
S2: leave-one-vehicle-and-map-out	raw profile	M2b RMS/v^0.5	6.88 [6.18, 7.68]	0.48 [0.41, 0.54]	0.07 [-0.07, 0.17]
S2: leave-one-vehicle-and-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.88 [6.18, 7.69]	0.47 [0.40, 0.52]	0.07 [-0.07, 0.16]
S2: leave-one-vehicle-and-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.70 [5.94, 7.57]	0.45 [0.36, 0.53]	0.09 [-0.04, 0.18]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (all features)	7.25 [6.64, 7.90]	0.46 [0.39, 0.51]	0.02 [-0.17, 0.15]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (no speed features)	7.39 [6.77, 8.05]	0.45 [0.39, 0.51]	-0.01 [-0.21, 0.12]
S2: leave-one-vehicle-and-map-out	raw profile	M5 1D-CNN	11.06 [8.13, 16.02]	0.43 [0.36, 0.49]	-173.32 [-605.45, -0.78]

TABLE S2
PVS: LEAVE-ONE-ROUTE-OUT LOGISTIC REGRESSIONS, AUC WITH CLUSTER-BOOTSTRAP 95% CIS OVER GPS CELLS.

Task	Predictors	AUC [95% CI]	windows	positives
A: asphalt vs cobblestone+dirt	speed	0.97 [0.94, 0.99]	1070	352
A: asphalt vs cobblestone+dirt	RMS	0.93 [0.91, 0.94]	1070	352
A: asphalt vs cobblestone+dirt	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model	0.94 [0.90, 0.96]	1070	352
A: asphalt vs cobblestone+dirt	model+speed	0.99 [0.97, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model+RMS	0.96 [0.95, 0.98]	1070	352
A: asphalt vs cobblestone+dirt	model+speed+RMS	0.99 [0.99, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	speed+RMS	0.99 [0.98, 1.00]	1070	352
B: good vs regular+bad	speed	0.97 [0.95, 0.99]	1070	358
B: good vs regular+bad	RMS	0.93 [0.91, 0.95]	1070	358
B: good vs regular+bad	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	358
B: good vs regular+bad	model	0.94 [0.90, 0.96]	1070	358
B: good vs regular+bad	model+speed	0.99 [0.97, 1.00]	1070	358
B: good vs regular+bad	model+RMS	0.97 [0.95, 0.98]	1070	358
B: good vs regular+bad	model+speed+RMS	1.00 [0.99, 1.00]	1070	358
B: good vs regular+bad	speed+RMS	0.99 [0.98, 1.00]	1070	358
C: bad vs regular (cobblestone+dirt only)	speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS	0.23 [0.15, 0.32]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS/sqrt(v)	0.43 [0.35, 0.51]	352	72
C: bad vs regular (cobblestone+dirt only)	model	0.41 [0.33, 0.50]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	model+RMS	0.51 [0.42, 0.58]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed+RMS	0.80 [0.72, 0.86]	352	72
C: bad vs regular (cobblestone+dirt only)	speed+RMS	0.78 [0.70, 0.84]	352	72

S3. SIMULATOR

The half-car is discretised exactly under a first-order hold and solved in modal coordinates; the solution deviates by at most $3.9\cdot10^{-13}$ (relative) from a direct state-space recursion and by $4.4\cdot10^{-9}$ from a fine-step Runge–Kutta integration. The sampled vehicles have body modes at 0.95–2.52 Hz and wheel-hop modes at 9.1–16.1 Hz. Speed is drawn independently of roughness (base speed 20–110 km/h with Ornstein–Uhlenbeck fluctuations and random braking events). Table S3 lists the parameter ranges; the variants and their seeds are listed in `evaluation/PROTOCOL_V2.md` and generated by `experiments/sim_generate.py`.

TABLE S3
PARAMETER RANGES OF THE PARAMETRIC SIMULATOR (SENSITIVITY RANGES; SOURCES IN CONFIG/SIM_RANGES.JSON).

group	parameter	range	unit	source
vehicle	m_s	1000 - 1800	kg	G, Z
vehicle	r_y	1 - 1.3	m	G (pitch radius of gyration ~ sqrt(a*b))
vehicle	L	2.4 - 3	m	G, Z
vehicle	a_frac	0.4 - 0.5	-	front-heavy passenger cars
vehicle	f_s	1 - 1.6	Hz	G, S
vehicle	rear_freq_ratio	1 - 1.2	-	G (flat-ride tuning)
vehicle	zeta	0.2 - 0.45	-	G, S
vehicle	m_u_per_wheel	30 - 50	kg	G
vehicle	k_t_per_tire	150000 - 300000	N/m	G (wheel hop 10-15 Hz)
vehicle	x_s	-0.5 - 1.2	m	sensor between CG and front axle (LiRA readme: near front axle)
sensor: autopi	rate / LSB / noise / tilt SD	55-62 Hz / 0.01 g / 150-400 ug/rtHz / 6.0 deg	-	M, D
sensor: phone	rate / LSB / noise / tilt SD	90-210 Hz / 0.0006 g / 100-300 ug/rtHz / 12.0 deg	-	D (smartphone IMUs); cradle resonance range is an assumption, treated as sensitivity
speed	base speed	20-110	km/h	design choice (independent of roughness)

S4. REAL CARS: ADDITIONAL RESULTS

Table S4 gives every estimator per held-out route. Table S7 gives paired differences at four levels of dependence: segment, 1 km and 5 km blocks, and routes; Fig. S3a shows why segment-level intervals are too narrow. Table S8 decomposes pass-level errors into segment, telematics-unit and residual components. Table S9 gives the unit-disjoint baseline, Table S10 the within-road anchor to Bajic et al., Table S11 the effect of averaging passes on a fixed set of segments, Table S12 the calibrated quarter-car inversion per fold (with the fitted transmissibility in Fig. S3b) and Table S13 the comparison of the two profilometers. Table S5 tests the within-route ranking of three trained estimators against the untrained statistic $\mathrm{RMS}/v^{0.5}$ route by route, and Table S6 gives relative errors by road class.

TABLE S4

PER-ROUTE RESULTS ON REAL CARS (SEGMENT MAE IN M/KM AND WITHIN-ROUTE SPEARMAN ρ; EACH HELD-OUT ROUTE EVALUATED BY THE MODELS TRAINED ON THE OTHER THREE; ρ IS UNDEFINED FOR PREDICTORS THAT ARE CONSTANT WITHIN A ROUTE). SEGMENTS: CPH1 509, CPH6 426, M3 515, M13 228. LAST ROWS: UNTRAINED VIBRATION STATISTICS, WHOSE WITHIN-ROUTE ρ NEEDS NO TRAINING DATA.

	MAE				ρ			
Estimator	CPH1	CPH6	M3	M13	CPH1	CPH6	M3	M13
M0 constant	2.12	2.03	1.31	0.79	–	–	–	–
M1 speed only	1.10	1.11	0.31	0.25	0.48	0.42	0.20	0.45
M2c vibration index	1.03	0.86	0.35	0.25	0.74	0.74	0.86	0.70
M3 LightGBM	0.60	0.70	0.13	0.15	0.88	0.84	0.90	0.72
M3 LightGBM, no speed	1.03	0.74	0.35	0.35	0.80	0.80	0.86	0.57
M4 MLP	0.60	0.70	0.13	0.16	0.88	0.83	0.89	0.72
M5 1D-CNN	0.73	0.68	0.15	0.17	0.87	0.79	0.88	0.71
Sim V0 LightGBM (zero-shot)	0.55	0.73	0.21	0.18	0.88	0.81	0.93	0.73
Sim V10 LightGBM (zero-shot)	0.57	0.69	0.26	0.21	0.88	0.81	0.93	0.73
M0b map-only road-class prior	1.20	1.18	0.34	0.30	–	–	–	–
M7 quarter-car inversion	0.68	0.71	0.27	0.17	0.86	0.82	0.89	0.72
M7 + output calibration	0.70	0.84	0.39	0.22	0.86	0.82	0.89	0.72
untrained: raw RMS	–	–	–	–	0.71	0.65	0.90	0.75
untrained: RMS/$v^{0.5}$	–	–	–	–	0.84	0.79	0.92	0.74
untrained: RMS/v	–	–	–	–	0.86	0.78	0.92	0.73

TABLE S5

PAIRED TEST OF WITHIN-ROUTE RANKING: SPEARMAN ρ (TOP) AND PARTIAL ρ GIVEN SEGMENT SPEED (BOTTOM) OF EACH ESTIMATOR MINUS THAT OF THE UNTRAINED STATISTIC RMS/$v^{0.5}$ ON THE SAME SEGMENTS, PER HELD-OUT ROUTE AND MEAN OVER ROUTES, WITH 95 % INTERVALS FROM 5 KM BLOCKS RESAMPLED WITHIN EACH ROUTE (B = 2000). M13 HAS ONLY 6 BLOCKS, SO ITS INTERVALS ARE COARSE.

	Estimator	CPH1	CPH6	M3	M13	mean
$\Delta\rho$	real-trained LightGBM	+0.042 [+0.027, +0.062]	+0.044 [+0.032, +0.066]	−0.020 [−0.034, −0.008]	−0.023 [−0.033, −0.010]	+0.011 [+0.004, +0.020]
	V0 LightGBM (zero-shot)	+0.038 [+0.016, +0.064]	+0.017 [+0.002, +0.040]	+0.007 [−0.005, +0.020]	−0.015 [−0.055, +0.023]	+0.012 [−0.001, +0.025]
	V1 LightGBM (zero-shot)	+0.021 [−0.003, +0.047]	+0.002 [−0.023, +0.027]	+0.004 [−0.018, +0.017]	−0.007 [−0.042, +0.036]	+0.005 [−0.008, +0.020]
Δ partial ρ	real-trained LightGBM	+0.026 [+0.019, +0.037]	+0.020 [+0.006, +0.037]	−0.020 [−0.032, −0.010]	−0.023 [−0.040, +0.007]	+0.001 [−0.006, +0.009]
	V0 LightGBM (zero-shot)	+0.019 [−0.004, +0.038]	−0.008 [−0.037, +0.020]	+0.006 [−0.003, +0.015]	−0.016 [−0.063, +0.028]	+0.000 [−0.014, +0.015]
	V1 LightGBM (zero-shot)	−0.001 [−0.022, +0.017]	−0.026 [−0.068, +0.007]	+0.000 [−0.013, +0.011]	−0.009 [−0.038, +0.044]	−0.009 [−0.023, +0.008]

TABLE S6

RELATIVE ERRORS ON REAL CARS: SEGMENT MAE OVER THE TWO HELD-OUT ROUTES OF EACH ROAD CLASS DIVIDED BY THEIR MEAN IRI (%; TABLE I ESTIMATORS, LEAVE-ONE-ROUTE-OUT).

Estimator	urban	motorway
M0 constant	62	100
M1 speed only	33	25
M2c vibration index	28	28
M3 LightGBM	19	12
M3 LightGBM, no speed	27	30
M4 MLP	19	12
M5 1D-CNN	21	14
Sim V0 LightGBM (zero-shot)	19	17
Sim V10 LightGBM (zero-shot)	19	22
M0b map-only road-class prior	35	28
M7 quarter-car inversion	21	21
M7 + output calibration	23	29

TABLE S7

PAIRED MAE DIFFERENCES (M/KM) AT FOUR LEVELS OF DEPENDENCE: CLUSTER BOOTSTRAP OVER SEGMENTS, 1 KM AND 5 KM BLOCKS (B = 2000), AND THE ROUTE-LEVEL DESCRIPTIVE INTERVAL (MEAN OF THE 4 PER-ROUTE DIFFERENCES, t WITH 3 DF). REFERENCE: REAL-TRAINED LIGHTGBM UNLESS STATED.

Estimator	Diff.	Segment	1 km block	5 km block	Route (t_3)
V0 LightGBM (zero-shot)	+0.019	[+0.008, +0.031]	[+0.002, +0.036]	[−0.003, +0.040]	[−0.06, +0.10]
V10 LightGBM (zero-shot)	+0.039	[+0.026, +0.052]	[+0.019, +0.059]	[+0.007, +0.068]	[−0.07, +0.15]
MLP (per-seed mean)	+0.005	[−0.003, +0.012]	[−0.007, +0.017]	[−0.012, +0.024]	[+0.00, +0.01]
MLP (ensemble)	+0.004	[−0.004, +0.011]	[−0.008, +0.016]	[−0.013, +0.023]	[+0.00, +0.01]
1D-CNN (per-seed mean)	+0.055	[+0.042, +0.068]	[+0.035, +0.076]	[+0.029, +0.086]	[−0.07, +0.16]
1D-CNN (ensemble)	+0.045	[+0.032, +0.059]	[+0.025, +0.066]	[+0.019, +0.075]	[−0.07, +0.15]
speed only	+0.324	[+0.295, +0.351]	[+0.277, +0.373]	[+0.252, +0.404]	[−0.01, +0.60]
vibration index M2c	+0.251	[+0.226, +0.275]	[+0.212, +0.292]	[+0.202, +0.306]	[−0.00, +0.46]
road-class prior (map)	+0.387	[+0.356, +0.418]	[+0.339, +0.438]	[+0.307, +0.475]	[+0.02, +0.70]
road-class prior (map) vs speed only	+0.064	[+0.045, +0.082]	[+0.028, +0.100]	[+0.029, +0.101]	[+0.02, +0.11]

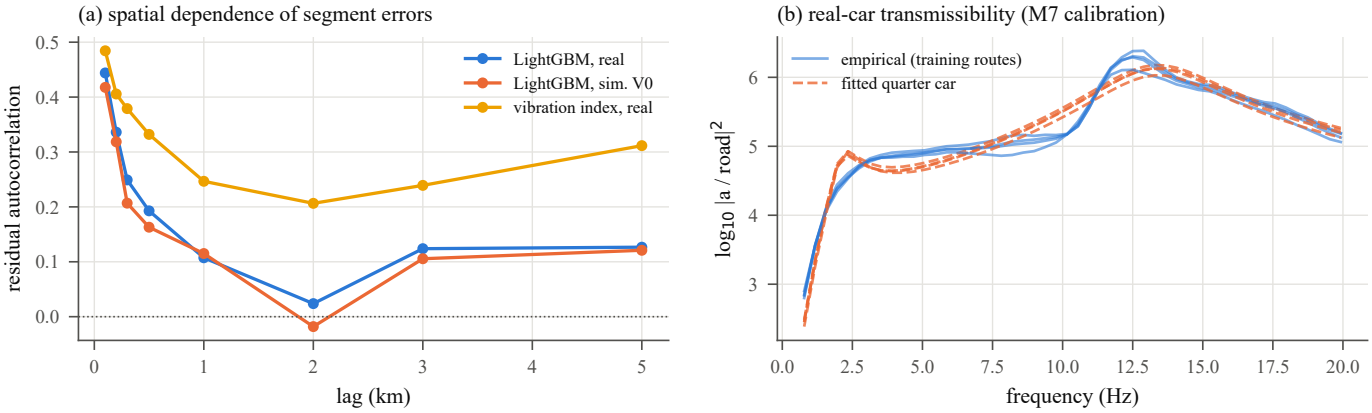


Fig. S3. (a) Autocorrelation of segment-level log errors along the road, by lag. (b) Empirical transmissibility of the real car (acceleration spectrum divided by the road displacement spectrum at the pass speed; training routes of each fold) and the fitted quarter car used by M7.

TABLE S8

VARIANCE COMPONENTS OF PASS-LEVEL LOG ERRORS (TWO-WAY DECOMPOSITION: SEGMENT EFFECTS + TELEMATICS-UNIT EFFECTS; SD IN % OF LOG ERROR; UNIT SD WITH A 1 KM-BLOCK BOOTSTRAP INTERVAL).

Model	Unit (vehicle) SD	Segment SD	Pass residual SD	Units
REAL GBM	7.3 [6.6, 8.5]	21.7	15.2	9
SIM P79	7.7 [6.8, 9.6]	21.8	16.2	9

TABLE S9

UNIT-DISJOINT REAL BASELINE: LIGHTGBM TRAINED ONLY ON PASSES OF TELEMATICS UNITS THAT NEVER DRIVE THE HELD-OUT ROUTE, VS A SIZE-MATCHED RANDOM SUBSET OF ALL UNITS (5 DRAWS; MEAN [MIN, MAX]) AND THE FULL V1 MODEL. THE M3 FOLD KEEPS ONLY URBAN TRAINING DATA AND IS NOT INTERPRETABLE AS A VEHICLE EFFECT.

Held-out route	Train units	Classes	Unit-disjoint	Size-matched	All units (v1)
CPH1	3,176	motorway, urban	0.63	0.63 [0.60, 0.65]	0.60
CPH6	11,976	motorway, urban	0.69	0.70 [0.69, 0.71]	0.70
M3	2,924	urban	0.62	0.16 [0.14, 0.19]	0.13
M13	7,323	motorway, urban	0.15	0.15 [0.15, 0.15]	0.15

TABLE S10

APPROXIMATE ANCHOR TO BAJIC ET AL. (2021): WITHIN-ROUTE SPATIAL HOLDOUT ON M3 (EACH DIRECTION: FIRST 80 % OF THE CHAINAGE FOR TRAINING, LAST 20 % FOR TESTING). BAJIC ET AL. REPORTED MAE 0.16–0.17, RMSE 0.21 AND R^2 0.64–0.66 FOR THEIR BEST MODELS (AND MAE 0.33 FOR THE MEAN BASELINE) WITH A DIFFERENT REFERENCE SURVEY AND PROCESSING (A PROFILER OPERATED BY THE DANISH ROAD DIRECTORATE, IRI PER 10 M AVERAGED OVER OVERLAPPING 100 M WINDOWS) AND THEIR OWN PASS SELECTION, SO THE NUMBERS ARE NOT DIRECTLY COMPARABLE.

Model	MAE	RMSE	R^2
train mean (baseline)	0.319	0.391	−0.02
LightGBM	0.115	0.171	0.81
vibration index (alpha fitted)	0.133	0.185	0.77
MLP (5-seed mean prediction)	0.119	0.175	0.80

TABLE S11
SEGMENT MAE VS NUMBER OF PASSES AVERAGED PER SEGMENT, ON THE FIXED SET OF SEGMENTS WITH $\geq$8 PASSES (20 RANDOM PASS SUBSETS).

Model	Road	1	2	3	5	8
real LightGBM	motorway	0.20	0.17	0.16	0.15	0.14
real LightGBM	urban	0.66	0.60	0.59	0.57	0.56
real index (M2c)	motorway	0.33	0.33	0.32	0.32	0.32
real index (M2c)	urban	1.19	1.10	1.06	1.03	1.00
sim LightGBM (V0)	motorway	0.24	0.22	0.21	0.20	0.20
sim LightGBM (V0)	urban	0.64	0.56	0.54	0.53	0.52

TABLE S12
CALIBRATED QUARTER-CAR SPECTRAL INVERSION (M7) PER HELD-OUT ROUTE: FITTED NORMALISED QUARTER CAR (FIT TO THE EMPIRICAL TRANSMISSIBILITY OF THE THREE TRAINING ROUTES) AND SEGMENT MAE WITHOUT AND WITH THE LOG-LINEAR OUTPUT CALIBRATION.

Route	K_1	K_2	C	U	gain	fit RMSE	MAE	MAE, calibrated
CPH1	1919	192	7.3	0.29	0.63	0.41	0.68	0.70
CPH6	1903	188	7.3	0.29	0.60	0.40	0.71	0.84
M3	1791	203	7.0	0.27	0.56	0.41	0.27	0.39
M13	1876	189	7.2	0.29	0.60	0.41	0.17	0.22

TABLE S13
ARAN9000 VS P79 (100 M, MEAN OF BOTH WHEEL PATHS) AFTER HEADING-CONSTRAINED MATCHING AND REMOVAL OF ZERO-IRI RECORDS. SURVEYS: P79 10-09-2020 (CPH1, M3, M13) AND 04-06-2020 (CPH6); ARAN 12-08-2020 (CPH1), 13-08-2020 (M3, M13) AND 30-08-2021 (CPH6).

Scope	Segments	MAE	Bias	Pearson	Spearman	Median ratio	$>$15 m/km (%)
all	1645	0.98	+0.85	0.68	0.96	1.10	2.7
CPH1	483	1.51	+1.30	0.60	0.87	1.08	3.9
CPH6	410	1.89	+1.63	0.50	0.80	1.06	6.1
M13	230	0.12	+0.09	0.96	0.93	1.10	0.0
M3	522	0.15	+0.14	0.95	0.94	1.12	0.0
urban	893	1.68	+1.45	0.54	0.84	1.07	4.9
motorway	752	0.14	+0.12	0.95	0.95	1.12	0.0

S5. Simulator variants: additional results

Tables S14 and S15 list all variants for LightGBM and for the vibration index trained in simulation, including the full V7 and V8 sets and the BeamNG corpus. Fig. S4 shows segment-level predictions of the real-trained model and of three variants. Table S16 compares every measured-profile variant with real-only training on the same k km, draw by draw, for the equal-budget design and the level offset, and Table S17 for the affine map and joint training. Table S18 gives intervals over the test routes for the level offset and joint training at 5 and 20 km, Table S19 the effect of keeping fewer passes per segment of the paired road, and Table S20 the value-of-simulation arms of our first analysis round, including the first joint-training results.

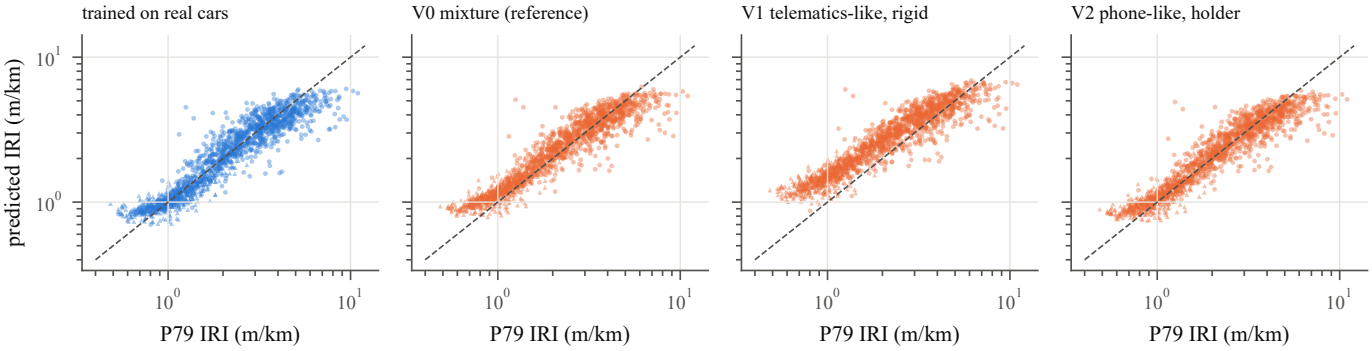


Fig. S4. Predicted vs P79 IRI on the held-out routes (segment means; circles urban, triangles motorway; log axes): LightGBM trained on real cars and trained only on simulator variants V0, V1 and V2.

TABLE S14

ALL SIMULATOR VARIANTS (LIGHTGBM), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE BEAMNG CORPUS WITH THE EARLIER (CORRECTED) AND RAW-PROFILE LABELS. COLUMNS AS IN TABLE II: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.44	−0.12	+0.17	0.83	0.69	−0.06	0.49	0.42 [0.40, 0.45]	0.42
V1 telematics-like, rigid	0.68	+0.32	+0.56	0.83	0.66	+0.02	0.59	0.53 [0.47, 0.60]	0.54
V7 telematics-like + holder	0.46	−0.05	+0.23	0.83	0.70	−0.15	0.50	0.43 [0.40, 0.46]	0.43
V8 phone-like, rigid	0.55	+0.21	+0.34	0.84	0.71	−0.00	0.51	0.45 [0.42, 0.49]	0.46
V2 phone-like + holder	0.41	−0.22	+0.06	0.83	0.69	−0.10	0.48	0.42 [0.40, 0.45]	0.41
V3 ideal sensor	0.67	+0.27	+0.56	0.83	0.62	+0.01	0.60	0.54 [0.48, 0.62]	0.55
V5 telematics matched to LiRA-CD	0.69	+0.25	+0.60	0.82	0.64	+0.01	0.62	0.56 [0.51, 0.64]	0.58
V6 matched telematics + phone-like	0.46	−0.24	+0.22	0.83	0.69	+0.06	0.52	0.45 [0.41, 0.48]	0.45
V4 one mid-range vehicle	0.63	+0.39	+0.41	0.83	0.66	−0.05	0.50	0.44 [0.41, 0.46]	0.43
V10 measured + synthetic profiles	0.46	−0.26	+0.23	0.84	0.66	−0.03	0.53	0.46 [0.41, 0.50]	0.46
V9 synthetic profiles only	0.71	+0.09	+0.70	0.81	0.66	+0.23	0.84	0.75 [0.65, 0.88]	0.77
V7, all passes	0.50	+0.00	+0.30	0.84	0.68	−0.11	0.52	0.45 [0.41, 0.49]	0.46
V8, all passes	0.55	+0.19	+0.32	0.83	0.70	+0.02	0.51	0.45 [0.42, 0.49]	0.46
BeamNG corpus, earlier labels	1.36	−2.21	−0.08	0.35	0.48	+0.79	1.77	1.59 [1.30, 1.87]	1.66
BeamNG corpus, raw-profile labels	1.97	−0.47	+2.94	0.32	0.42	+1.39	2.56	2.22 [1.66, 2.85]	2.42
trained on real cars (M3)	0.42	−0.19	−0.02	0.83	0.66	−0.15	–	–	–

TABLE S15

ALL SIMULATOR VARIANTS (VIBRATION INDEX WITH THE SPEED EXPONENT FITTED IN SIMULATION), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE BEAMNG CORPUS WITH THE EARLIER (CORRECTED) AND RAW-PROFILE LABELS. COLUMNS AS IN TABLE II: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.57	+0.24	+0.28	0.81	0.75	−0.48	0.57	0.52 [0.48, 0.55]	0.52
V1 telematics-like, rigid	0.66	+0.53	+0.36	0.81	0.76	−0.49	0.56	0.51 [0.47, 0.54]	0.51
V7 telematics-like + holder	0.56	+0.24	+0.28	0.82	0.75	−0.45	0.56	0.51 [0.47, 0.55]	0.51
V8 phone-like, rigid	0.63	+0.44	+0.34	0.81	0.76	−0.47	0.58	0.52 [0.48, 0.56]	0.52
V2 phone-like + holder	0.55	+0.13	+0.21	0.81	0.77	−0.49	0.58	0.52 [0.49, 0.56]	0.53
V3 ideal sensor	0.63	+0.38	+0.40	0.81	0.72	−0.45	0.58	0.52 [0.48, 0.57]	0.53
V5 telematics matched to LiRA-CD	0.63	+0.33	+0.42	0.82	0.73	−0.42	0.58	0.52 [0.47, 0.57]	0.53
V6 matched telematics + phone-like	0.54	+0.05	+0.33	0.82	0.72	−0.40	0.59	0.53 [0.48, 0.58]	0.53
V4 one mid-range vehicle	0.74	+0.68	+0.42	0.82	0.78	−0.45	0.57	0.51 [0.47, 0.55]	0.52
V10 measured + synthetic profiles	0.57	+0.18	+0.36	0.82	0.74	−0.39	0.59	0.53 [0.47, 0.58]	0.53
V9 synthetic profiles only	0.69	−0.20	+0.74	0.83	0.65	+0.05	0.98	0.92 [0.78, 1.07]	0.95
V7, all passes	0.55	+0.12	+0.31	0.82	0.73	−0.41	0.58	0.53 [0.48, 0.58]	0.53
V8, all passes	0.63	+0.44	+0.33	0.81	0.77	−0.47	0.58	0.52 [0.48, 0.55]	0.52
BeamNG corpus, earlier labels	0.85	−1.37	+0.08	0.83	0.50	−0.11	1.04	0.97 [0.85, 1.14]	1.00
BeamNG corpus, raw-profile labels	1.96	+1.51	+2.36	0.82	0.46	+0.08	1.25	1.18 [1.02, 1.40]	1.23
trained on real cars (M3)	0.67	−0.44	+0.26	0.76	0.50	−0.64	–	–	–

TABLE S16

EVERY MEASURED-PROFILE SIMULATOR VARIANT AGAINST LIGHTGBM TRAINED ON THE REAL PASSES OVER THE SAME k KM (20 PAIRED DRAWS OF TRAINING-ROUTE BLOCKS). ENTRIES: MEAN DIFFERENCE IN POOLED SEGMENT MAE, SIMULATION ARM MINUS REAL-ONLY (M/KM; NEGATIVE = SIMULATION BETTER), AND IN PARENTHESES THE NUMBER OF THE 20 DRAWS IN WHICH THE SIMULATION ARM WAS BETTER. "SAME k KM": TRAINED ON SIMULATED PASSES OVER THE PROFILES OF THE SAME BLOCKS ONLY (EQUAL PROFILE BUDGET; V1–V8 ADDED AFTER THE V0 RESULT WAS KNOWN, CLASS E). "ALL PROFILES + LEVEL": TRAINED ON ALL TRAINING-ROUTE PROFILES, LEVEL OFFSET FROM THE k KM OF REAL PASSES. "+ AFFINE": AFFINE CALIBRATION INSTEAD OF THE OFFSET.

	Same k km of profiles			All profiles + level from k km				+ affine	
Variant	2 km	5 km	20 km	1 km	5 km	10 km	20 km	5 km	20 km
V0	−0.28 (20)	−0.08 (16)	+0.01 (9)	−1.14 (20)	−0.25 (20)	−0.10 (20)	−0.04 (19)	−0.20 (19)	−0.03 (17)
V1	−0.09 (16)	+0.11 (2)	+0.24 (0)	−1.04 (20)	−0.15 (14)	+0.02 (5)	+0.08 (0)	−0.18 (18)	+0.02 (7)
V2	−0.27 (20)	−0.08 (16)	−0.02 (17)	−1.15 (20)	−0.26 (20)	−0.11 (20)	−0.05 (20)	−0.21 (19)	−0.03 (19)
V3	−0.16 (18)	+0.06 (4)	+0.22 (0)	−1.03 (20)	−0.14 (14)	+0.04 (4)	+0.09 (0)	−0.20 (19)	−0.01 (12)
V4	−0.17 (18)	+0.03 (6)	+0.19 (0)	−1.13 (20)	−0.24 (20)	−0.09 (20)	−0.03 (18)	−0.21 (20)	−0.02 (17)
V5	−0.12 (16)	+0.09 (3)	+0.23 (0)	−1.01 (20)	−0.12 (13)	+0.06 (3)	+0.12 (0)	−0.18 (18)	+0.02 (5)
V6	−0.24 (20)	−0.05 (13)	+0.02 (4)	−1.11 (20)	−0.23 (20)	−0.07 (17)	−0.01 (14)	−0.21 (19)	−0.02 (16)
V7	−0.22 (19)	−0.03 (10)	+0.05 (0)	−1.13 (20)	−0.25 (20)	−0.09 (20)	−0.03 (18)	−0.20 (18)	−0.01 (13)
V8	−0.21 (18)	−0.01 (9)	+0.10 (0)	−1.12 (20)	−0.23 (20)	−0.06 (18)	−0.00 (14)	−0.20 (19)	−0.02 (14)
Real-only MAE	1.14	0.68	0.46	1.63	0.68	0.52	0.46	0.68	0.46

TABLE S17

EVERY SIMULATOR VARIANT COMBINED WITH k KM OF PAIRED DRIVES THROUGH AN AFFINE MAP (FITTED ON THE PAIRED PASSES) OR BY JOINT TRAINING (SIMULATED UNITS AND PAIRED PASSES, EQUAL TOTAL WEIGHT), AGAINST REAL-ONLY LIGHTGBM ON THE SAME PASSES (20 PAIRED DRAWS). ENTRIES: MEAN DIFFERENCE IN POOLED SEGMENT MAE (M/KM; NEGATIVE = SIMULATION BETTER) AND, IN PARENTHESES, THE NUMBER OF DRAWS IN WHICH SIMULATION WAS BETTER. LAST ROW: POOLED MAE OF A NO-SIMULATOR INDEX $\log \mathrm{IRI} = c + b \log(\mathrm{RMS}/v^{\alpha})$ WITH α THE CAR'S LABEL-FREE WITHIN-SEGMENT SPEED EXPONENT (0.59–0.72), AFFINE-CALIBRATED ON THE SAME PASSES.

	+ affine map				+ joint training			
Variant	1 km	5 km	10 km	20 km	1 km	5 km	10 km	20 km
V0	−0.42 (17)	−0.20 (19)	−0.07 (19)	−0.03 (17)	−1.17 (20)	−0.23 (20)	−0.07 (19)	−0.03 (20)
V1	−0.45 (18)	−0.18 (18)	−0.03 (11)	+0.02 (7)	−1.08 (20)	−0.21 (20)	−0.06 (18)	−0.02 (17)
V2	−0.42 (18)	−0.21 (19)	−0.08 (19)	−0.03 (19)	−1.19 (20)	−0.23 (20)	−0.08 (20)	−0.03 (20)
V3	−0.25 (18)	−0.20 (19)	−0.05 (14)	−0.01 (12)	−1.07 (20)	−0.21 (20)	−0.05 (18)	−0.01 (16)
V4	−0.71 (19)	−0.21 (20)	−0.07 (19)	−0.02 (17)	−1.08 (20)	−0.21 (20)	−0.06 (19)	−0.02 (18)
V5	−0.60 (18)	−0.18 (18)	−0.02 (11)	+0.02 (5)	−1.07 (20)	−0.21 (20)	−0.05 (19)	−0.02 (18)
V6	−0.61 (18)	−0.21 (19)	−0.07 (17)	−0.02 (16)	−1.15 (20)	−0.22 (20)	−0.07 (19)	−0.03 (19)
V7	−0.55 (18)	−0.20 (18)	−0.05 (16)	−0.01 (13)	−1.16 (20)	−0.22 (20)	−0.06 (19)	−0.02 (18)
V8	−0.46 (18)	−0.20 (19)	−0.06 (17)	−0.02 (14)	−1.13 (20)	−0.22 (20)	−0.06 (19)	−0.02 (18)
V10	−0.41 (18)	−0.20 (19)	−0.08 (19)	−0.03 (17)	−1.15 (20)	−0.23 (20)	−0.07 (19)	−0.03 (20)
Real-only LightGBM (MAE)	1.63	0.68	0.52	0.46	1.63	0.68	0.52	0.46
No-simulator index (MAE)	1.09	0.78	0.82	0.76				

TABLE S18

TEST-ROUTE UNCERTAINTY OF THE CALIBRATION COMPARISON: MEAN OVER THE 20 DRAWS OF THE POOLED-MAE DIFFERENCE TO REAL-ONLY LIGHTGBM ON THE SAME k KM (M/KM), WITH A 95 % INTERVAL FROM A BOOTSTRAP OVER 5 KM BLOCKS OF THE TEST SEGMENTS (40 BLOCKS). THE INTERVAL CONDITIONS ON THE 20 CALIBRATION DRAWS.

	+ level offset		+ joint training	
Variant	5 km	20 km	5 km	20 km
V0	−0.25 [−0.29, −0.22]	−0.04 [−0.05, −0.03]	−0.23 [−0.26, −0.19]	−0.03 [−0.03, −0.02]
V1	−0.15 [−0.18, −0.12]	+0.08 [+0.04, +0.11]	−0.21 [−0.26, −0.17]	−0.02 [−0.03, −0.01]
V2	−0.26 [−0.30, −0.22]	−0.05 [−0.06, −0.03]	−0.23 [−0.26, −0.20]	−0.03 [−0.04, −0.02]
V3	−0.14 [−0.17, −0.11]	+0.09 [+0.05, +0.13]	−0.21 [−0.25, −0.16]	−0.01 [−0.03, −0.00]
V4	−0.24 [−0.28, −0.21]	−0.03 [−0.05, −0.01]	−0.21 [−0.26, −0.16]	−0.02 [−0.03, −0.01]
V5	−0.12 [−0.15, −0.09]	+0.12 [+0.08, +0.16]	−0.21 [−0.26, −0.17]	−0.02 [−0.03, −0.01]
V6	−0.23 [−0.26, −0.20]	−0.01 [−0.04, +0.01]	−0.22 [−0.26, −0.19]	−0.03 [−0.04, −0.02]
V7	−0.25 [−0.29, −0.21]	−0.03 [−0.04, −0.01]	−0.22 [−0.26, −0.18]	−0.02 [−0.03, −0.01]
V8	−0.23 [−0.25, −0.20]	−0.00 [−0.02, +0.01]	−0.22 [−0.27, −0.18]	−0.02 [−0.03, −0.01]

TABLE S19

PASS DENSITY OF THE PAIRED DATA AT $k = 5$ KM: EACH SEGMENT KEEPS AT MOST m OF ITS REAL PASSES (RANDOM, INDEPENDENT OF THE LABELS; 20 DRAWS). POOLED MAE IN M/KM; FOR THE SIMULATION ARMS THE RANGE OVER V0–V8 AND, IN PARENTHESES, THE RANGE OVER THE VARIANTS OF THE NUMBER OF DRAWS IN WHICH THE ARM BEAT REAL-ONLY LIGHTGBM WITH THE SAME PASSES.

m	segment-passes	real-only LightGBM	+ level offset	+ joint training
1	49	0.93	0.43–0.55 (20)	0.44–0.49 (20)
2	98	0.75	0.42–0.55 (18–20)	0.44–0.48 (20)
5	242	0.69	0.42–0.55 (13–20)	0.45–0.48 (20)
all	475	0.68	0.42–0.56 (13–20)	0.45–0.47 (20)

TABLE S20

FIRST-ROUND VALUE-OF-SIMULATION ARMS AS SPECIFIED IN OUR FIRST PROTOCOL AND RUN IN THE FIRST ANALYSIS ROUND (5 DRAWS PER k, THE V10 SIMULATION SET, POOLED SEGMENT MAE IN M/KM, MEAN OVER DRAWS; THE CNNS ARE SEED-AVERAGED AND DIVERGE ON SOME DRAWS AT SMALL k). SHOWN FOR COMPLETENESS; THE MAIN ANALYSIS USES 20 DRAWS AND EVERY SIMULATOR VARIANT.

Arm	1	2	5	10	20	all
real-only 1D-CNN	>50	–	1.25	–	0.57	0.55
real-only LightGBM	1.41	1.18	0.82	0.50	0.46	0.42
real-only physics index	1.10	0.93	0.81	0.71	0.67	0.67
sim physics index + real intercept	0.53	0.52	0.54	0.54	0.54	0.53
sim+real LightGBM (balanced weights)	0.46	0.46	0.45	0.44	0.43	0.41
sim-pretrained 1D-CNN + real fine-tuning	>50	–	0.78	–	0.53	0.56

S6. Spectra and speed: additional results

Table S21 gives the band means of the spectral differences in Fig. 3a, Table S22 the 10–15 Hz ratio by road class and the speed scaling of raw vibration in simulation and reality, Table S23 the association of every band's amplitude ratio with the zero-shot bias and level, Table S24 the band-limited retraining, Fig. S5 and Table S25 the fixed-road speed elasticities of all estimators with paired differences to the real-trained estimators, and Table S26 their robustness to unit fixed effects and to the within-pass speed variability. In the 333 Hz passes, 25 % of the vibration energy above 0.3 Hz lies at 10–20 Hz, 21 % above 20 Hz and 13 % above the 26.3 Hz Nyquist frequency of the slower logging (medians over 5,721 units).

TABLE S21

Mean $\log_{10}$ PSD difference by band, real minus simulated (re-simulated passes of the first 10 training vehicles over all P79 profiles), on common segment × 10 km/h cells; last row: real passes logged at 52.6 Hz minus 333 Hz (all units, device-confounded).

Comparison	0.5–2 Hz	2–5 Hz	5–10 Hz	10–15 Hz	15–20 Hz	Cells
all units: 52.6Hz-333Hz	−0.06	−0.09	−0.16	−0.24	−0.40	1,765
real-V1	−0.18	+0.38	+0.01	+0.69	+0.89	3,577
real-V2	−0.20	+0.39	−0.14	+0.27	+0.06	3,716
real-V3	−0.19	+0.49	+0.08	+0.71	+0.97	3,455
real-V5	−0.20	+0.55	+0.12	+0.69	+0.91	3,127
real-V7	−0.18	+0.43	−0.01	+0.47	+0.48	3,574
real-V8	−0.15	+0.37	−0.05	+0.45	+0.48	3,277

TABLE S22

Simulated vs real vibration on the same segments, per variant. Left: real/simulated amplitude ratio of the 10–15 Hz band (cell means over segment × 10 km/h speed band; all routes, urban, motorway) next to the zero-shot bias. Right: speed scaling of the raw vibration RMS on a fixed road (within-segment elasticity; simulated passes restricted to the speed range of the real passes over the same segment; 95 % CI from a 5 km-block bootstrap).

	10–15 Hz ratio real/sim			Zero-shot	RMS speed elasticity	
Source	all	urban	motorway	bias	urban	motorway
V0	1.86	1.57	2.14	+0.01	1.25 [1.17, 1.34]	1.10 [1.02, 1.18]
V1	2.41	1.98	2.81	+0.43	1.22 [1.13, 1.31]	1.07 [1.01, 1.17]
V2	1.36	1.33	1.39	−0.09	1.30 [1.19, 1.42]	1.29 [1.20, 1.34]
V3	2.37	1.92	2.81	+0.40	1.16 [1.10, 1.23]	1.40 [1.31, 1.46]
V4	1.93	1.67	2.19	+0.40	1.11 [1.07, 1.15]	1.26 [1.12, 1.41]
V5	2.26	1.93	2.60	+0.40	0.92 [0.84, 1.01]	1.32 [1.21, 1.41]
V6	1.80	1.49	2.12	−0.04	0.95 [0.85, 1.06]	1.17 [1.07, 1.25]
V7	1.80	1.57	2.00	+0.07	1.27 [1.14, 1.39]	1.27 [1.12, 1.35]
V8	1.89	1.60	2.15	+0.27	1.05 [0.97, 1.10]	1.21 [1.08, 1.31]
Real car	1	1	1	–	0.60 [0.50, 0.69]	0.85 [0.74, 0.93]

TABLE S23

Association, over the nine measured-profile variants V0–V8, between the log real/simulated amplitude ratio of each vibration feature (same segments and speed bands) and the zero-shot pooled bias or the mean log level: Spearman and Pearson correlations and R^2 of a linear fit. The variants are not independent (V0 mixes V1 and V2; V6 shares V2's phone-like sensor).

	vs bias (m/km)			vs mean log level		
Feature	Spearman	Pearson	R^2	Spearman	Pearson	R^2
0.5–2 Hz	0.22	0.32	0.10	0.08	0.22	0.05
2–5 Hz	0.25	0.44	0.19	0.33	0.41	0.17
5–10 Hz	0.80	0.81	0.66	0.80	0.86	0.73
10–15 Hz	0.97	0.84	0.71	0.90	0.90	0.81
overall RMS	0.88	0.88	0.78	0.83	0.88	0.77

TABLE S24

BAND-LIMITED ZERO-SHOT RETRAINING (LIGHTGBM, FIRST CONFIGURATION) FOR V0–V8 WITH THREE FEATURE SETS: ALL 13 FEATURES; ALL BUT THE 10–15 HZ BAND; ONLY FEATURES WITHOUT CONTENT ABOVE 10 HZ (LOG MEAN SPEED, SPEED VARIATION, BAND RMS 0.5–2, 2–5, 5–10 HZ). LEVEL: MEAN LOG RATIO BY ROAD CLASS; POOLED MAE ZERO-SHOT AND AFTER A LEVEL OFFSET FROM 5 KM OF PAIRED DRIVES (20 DRAWS); WITHIN-ROUTE ρ. LAST ROWS: SPREAD (MAX − MIN) OVER THE EIGHT SENSOR VARIANTS (ALL BUT V4).

	all 13					all but 10–15 Hz					below 10 Hz				
Variant	lvl urb.	lvl mot.	MAE	+offset	ρ	lvl urb.	lvl mot.	MAE	+offset	ρ	lvl urb.	lvl mot.	MAE	+offset	ρ
V0	+0.01	+0.14	0.44	0.42	0.83	+0.01	+0.16	0.44	0.43	0.83	−0.09	+0.14	0.49	0.51	0.80
V1	+0.14	+0.40	0.68	0.53	0.83	+0.14	+0.40	0.69	0.53	0.83	−0.10	+0.19	0.51	0.55	0.81
V2	−0.03	+0.06	0.41	0.42	0.83	−0.02	+0.07	0.41	0.42	0.83	−0.05	+0.13	0.48	0.49	0.80
V3	+0.13	+0.40	0.67	0.54	0.83	+0.13	+0.40	0.67	0.54	0.83	−0.04	+0.24	0.53	0.56	0.80
V4	+0.16	+0.29	0.63	0.44	0.83	+0.16	+0.31	0.64	0.44	0.83	+0.10	+0.33	0.62	0.53	0.80
V5	+0.12	+0.42	0.69	0.56	0.82	+0.13	+0.43	0.70	0.56	0.82	−0.09	+0.24	0.53	0.57	0.80
V6	−0.03	+0.19	0.46	0.45	0.83	−0.03	+0.20	0.46	0.45	0.83	−0.09	+0.20	0.53	0.55	0.79
V7	+0.02	+0.18	0.46	0.43	0.83	+0.03	+0.21	0.48	0.43	0.83	−0.12	+0.18	0.52	0.53	0.80
V8	+0.10	+0.26	0.55	0.45	0.84	+0.11	+0.27	0.56	0.45	0.83	−0.08	+0.12	0.47	0.48	0.80
spread, 8 sensor variants	0.17	0.36				0.17	0.36				0.08	0.12			

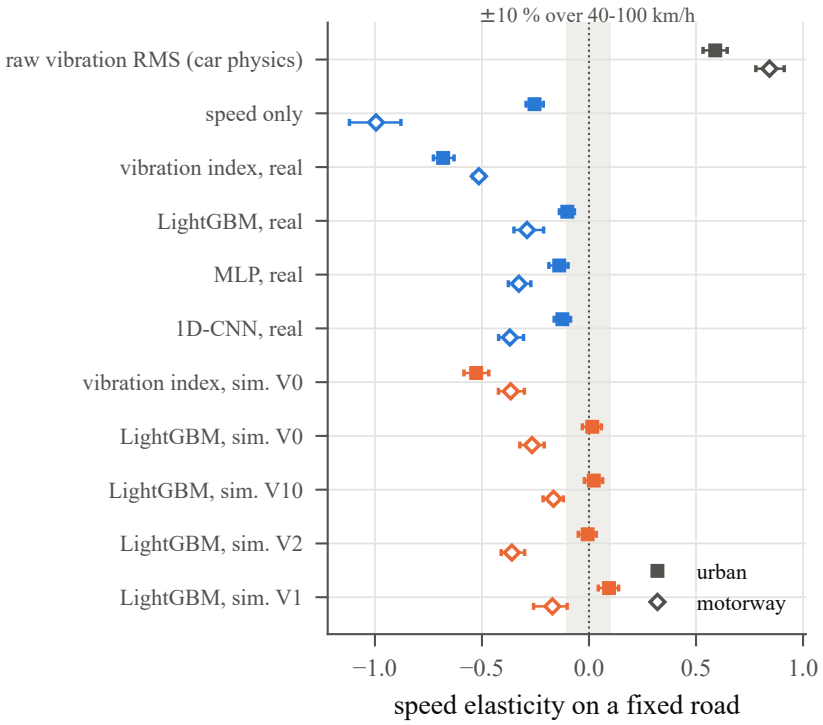


Fig. S5. Speed elasticity of each estimate on a fixed real road (slope of log estimate on log speed with segment fixed effects; 95 % intervals from a 1 km-block bootstrap; urban and motorway segments). Grey band: ±10 % change between 40 and 100 km/h. Top row: raw vibration RMS of the real car (main paper, Section V-E).

TABLE S25

SPEED ELASTICITY OF EACH ESTIMATE ON A FIXED ROAD: SLOPE OF LOG(PREDICTION) ON LOG(SPEED) WITH SEGMENT FIXED EFFECTS (SEGMENTS WITH $\geq$3 PASSES), 95 % CI BY 1 KM-BLOCK BOOTSTRAP; "BETWEEN": DESIGN WITH ROUTE EFFECTS AND LOG IRI AS COVARIATE. |ELASTICITY| $\leq$0.104 CORRESPONDS TO THE $\pm$10 % CRITERION OVER 40–100 KM/H. LAST COLUMN: PAIRED DIFFERENCE TO THE REAL-TRAINED ESTIMATOR OF THE SAME FAMILY (LIGHTGBM OR INDEX) ON THE SAME PASSES, 95 % CI BY 5 KM-BLOCK BOOTSTRAP (POOLED).

Estimate	Pooled	Urban	Motorway	Between	− real-trained
RAW RMS	+0.66 [+0.60, +0.71]	+0.59 [+0.53, +0.65]	+0.84 [+0.78, +0.91]	+0.64	–
REAL SPEED	−0.46 [−0.55, −0.37]	−0.25 [−0.29, −0.21]	−0.99 [−1.12, −0.88]	−0.46	–
REAL INDEX	−0.64 [−0.68, −0.59]	−0.68 [−0.73, −0.63]	−0.51 [−0.54, −0.49]	−0.65	–
REAL RMS index	+0.21 [+0.18, +0.24]	+0.20 [+0.16, +0.24]	+0.26 [+0.24, +0.28]	+0.21	–
REAL GBM	−0.15 [−0.19, −0.11]	−0.10 [−0.14, −0.07]	−0.29 [−0.35, −0.21]	−0.16	–
REAL GBM nospeed	−0.14 [−0.18, −0.10]	−0.21 [−0.24, −0.18]	+0.02 [−0.07, +0.18]	−0.17	–
REAL MLP	−0.19 [−0.23, −0.15]	−0.14 [−0.19, −0.10]	−0.33 [−0.38, −0.27]	−0.19	–
REAL CNN	−0.19 [−0.24, −0.15]	−0.12 [−0.16, −0.09]	−0.37 [−0.42, −0.31]	−0.19	–
SIM V0 GBM	−0.06 [−0.11, −0.01]	+0.02 [−0.03, +0.06]	−0.27 [−0.32, −0.21]	−0.08	+0.09 [+0.06, +0.12]
SIM V0 INDEX	−0.48 [−0.53, −0.44]	−0.53 [−0.58, −0.47]	−0.37 [−0.42, −0.30]	−0.50	+0.15 [+0.12, +0.18]
SIM V1 GBM	+0.02 [−0.03, +0.07]	+0.09 [+0.04, +0.14]	−0.17 [−0.26, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V1 INDEX	−0.49 [−0.54, −0.44]	−0.54 [−0.59, −0.48]	−0.36 [−0.42, −0.30]	−0.51	+0.15 [+0.11, +0.17]
SIM V2 GBM	−0.10 [−0.16, −0.05]	−0.01 [−0.05, +0.03]	−0.36 [−0.41, −0.30]	−0.11	+0.05 [+0.01, +0.08]
SIM V2 INDEX	−0.49 [−0.55, −0.44]	−0.55 [−0.61, −0.49]	−0.35 [−0.40, −0.28]	−0.52	+0.14 [+0.11, +0.17]
SIM V3 GBM	+0.01 [−0.04, +0.05]	+0.09 [+0.05, +0.13]	−0.22 [−0.29, −0.16]	−0.02	+0.16 [+0.12, +0.19]
SIM V3 INDEX	−0.45 [−0.50, −0.40]	−0.51 [−0.56, −0.45]	−0.30 [−0.36, −0.24]	−0.48	+0.19 [+0.15, +0.21]
SIM V4 GBM	−0.05 [−0.10, −0.00]	+0.03 [−0.01, +0.07]	−0.26 [−0.32, −0.21]	−0.05	+0.10 [+0.08, +0.13]
SIM V4 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.30 [−0.36, −0.24]	−0.47	+0.19 [+0.14, +0.22]
SIM V5 GBM	+0.01 [−0.03, +0.05]	+0.08 [+0.04, +0.12]	−0.16 [−0.25, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V5 INDEX	−0.42 [−0.47, −0.37]	−0.48 [−0.54, −0.43]	−0.24 [−0.30, −0.18]	−0.44	+0.22 [+0.19, +0.25]
SIM V6 GBM	+0.06 [+0.01, +0.11]	+0.16 [+0.11, +0.20]	−0.19 [−0.27, −0.12]	+0.05	+0.21 [+0.18, +0.24]
SIM V6 INDEX	−0.40 [−0.45, −0.35]	−0.45 [−0.50, −0.39]	−0.26 [−0.31, −0.19]	−0.42	+0.24 [+0.21, +0.26]
SIM V7 GBM	−0.15 [−0.20, −0.11]	−0.10 [−0.16, −0.06]	−0.28 [−0.33, −0.22]	−0.18	+0.00 [−0.04, +0.03]
SIM V7 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.33 [−0.39, −0.26]	−0.47	+0.18 [+0.15, +0.21]
SIM V8 GBM	−0.00 [−0.05, +0.05]	+0.08 [+0.03, +0.13]	−0.22 [−0.28, −0.15]	−0.01	+0.15 [+0.11, +0.18]
SIM V8 INDEX	−0.47 [−0.52, −0.42]	−0.52 [−0.58, −0.46]	−0.33 [−0.39, −0.26]	−0.49	+0.17 [+0.13, +0.20]
SIM V9 GBM	+0.23 [+0.19, +0.27]	+0.24 [+0.19, +0.28]	+0.21 [+0.16, +0.27]	+0.22	+0.38 [+0.33, +0.43]
SIM V9 INDEX	+0.05 [−0.01, +0.10]	−0.01 [−0.07, +0.04]	+0.21 [+0.15, +0.27]	+0.02	+0.68 [+0.65, +0.71]
SIM V10 GBM	−0.03 [−0.07, +0.01]	+0.02 [−0.02, +0.06]	−0.17 [−0.21, −0.12]	−0.05	+0.12 [+0.10, +0.15]
SIM V10 INDEX	−0.39 [−0.44, −0.34]	−0.44 [−0.50, −0.38]	−0.25 [−0.31, −0.19]	−0.41	+0.25 [+0.21, +0.27]
SIM B corr GBM	+0.79 [+0.67, +0.91]	+0.76 [+0.60, +0.91]	+0.88 [+0.75, +0.98]	+0.65	+0.94 [+0.77, +1.11]
SIM B corr INDEX	−0.11 [−0.15, −0.08]	−0.16 [−0.20, −0.12]	+0.00 [−0.04, +0.05]	−0.13	+0.52 [+0.50, +0.54]

TABLE S26

ROBUSTNESS OF THE FIXED-ROAD SPEED ELASTICITIES: (A) DESIGN OF THE MAIN PAPER; (B) WITH TELEMATICS-UNIT FIXED EFFECTS (POINT ESTIMATE); (C) WITH THE LOG WITHIN-PASS SPEED SD AS A COVARIATE; (D) WITH THE SPEED COEFFICIENT OF VARIATION AS A COVARIATE, WHICH IS MECHANICALLY TIED TO $1/v$ (WITHIN-SEGMENT CORRELATION WITH LOG SPEED -0.86 ON URBAN ROADS) AND ABSORBS MOST OF THE WITHIN-SEGMENT SPEED VARIATION. 95 % INTERVALS FROM 5 KM BLOCKS. WITHIN URBAN SEGMENTS, LOG SPEED AND LOG SPEED SD CORRELATE AT -0.50: SLOW URBAN PASSES ARE MORE DYNAMIC.

Class	Quantity	(a) main design	(b) + unit FE	(c) + log speed SD	(d) + speed CV
urban	raw vibration RMS	+0.60 [+0.51, +0.69]	+0.59	+0.65 [+0.49, +0.84]	+0.83 [+0.47, +1.40]
	real-trained LightGBM	−0.10 [−0.17, −0.05]	−0.10	−0.06 [−0.16, +0.02]	−0.08 [−0.23, +0.13]
	V0 LightGBM (zero-shot)	+0.02 [−0.05, +0.08]	+0.02	+0.06 [−0.07, +0.17]	−0.15 [−0.28, +0.03]
	V0 minus real-trained	+0.12 [+0.09, +0.15]	+0.12	+0.12 [+0.09, +0.16]	−0.07 [−0.21, +0.01]
motorway	raw vibration RMS	+0.85 [+0.73, +0.92]	+0.83	+0.86 [+0.74, +0.96]	+0.91 [+0.75, +1.03]
	real-trained LightGBM	−0.29 [−0.34, −0.14]	−0.31	−0.24 [−0.30, −0.14]	−0.18 [−0.27, −0.02]
	V0 LightGBM (zero-shot)	−0.27 [−0.32, −0.21]	−0.27	−0.22 [−0.28, −0.17]	−0.23 [−0.29, −0.13]
	V0 minus real-trained	+0.02 [−0.08, +0.06]	+0.04	+0.01 [−0.07, +0.05]	−0.05 [−0.14, +0.00]

S7. Controlled simulation: other vehicles and sensors

LiRA-CD contains one car model, so transfer to other vehicles was examined only in simulation (class S, freshly simulated test data; Table S27, Fig. S6). With the amount of training data fixed at 1,200 passes, the MAE on 30 unseen vehicles exceeds that on the training vehicles by +70 % with 3 training vehicles, +16 % with 30 (the number used in V0) and +6 % with 100. The criterion of our earlier study (upper bound ≤10 %) is met from 100 vehicles with a segment bootstrap but not with a route bootstrap (upper bounds 11 and 12 % for 100 and 300 vehicles). The gap closes from both sides: the error on unseen vehicles falls (1.06 to 0.87 m/km from 3 to 100 vehicles) and the error on training vehicles rises as each gets fewer passes (0.62 to 0.82 m/km). At $N = 30$, a vehicle's error is only weakly related to its parameter distance from the nearest training vehicle (Spearman 0.14 over 30 vehicles). Sensor perturbations at test time (Table S27, right) reproduce in simulation what the variants showed on the real car: the sensor path moves the level. A holder resonance at 15 Hz turns a bias of −0.65 into +0.37 m/km; 25 Hz sampling and a speed error of +20 % are the most harmful perturbations (MAE 1.39 and 1.06 m/km), white noise, tilt and lost samples the least. Table S28 lists the controlled tests of our first analysis round (unseen vehicles, sensors and speeds on synthetic routes), and Fig. S7 the sensor-robustness experiment summarised in Table S27.

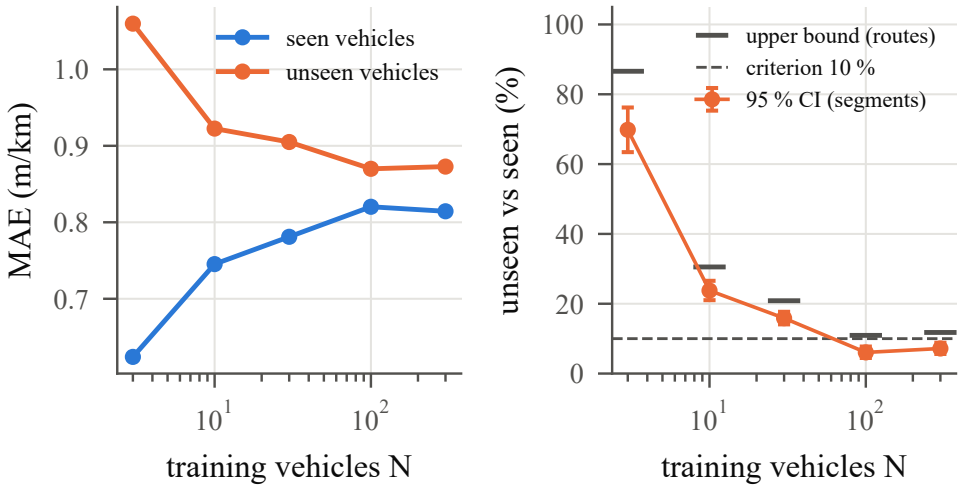


Fig. S6. Number of training vehicles at a fixed 1,200 training passes (simulation only). Left: MAE on seen and unseen vehicles. Right: relative increase from seen to unseen vehicles with 95 % interval (segment bootstrap) and the upper bound of a route bootstrap; dashed: 10 % criterion.

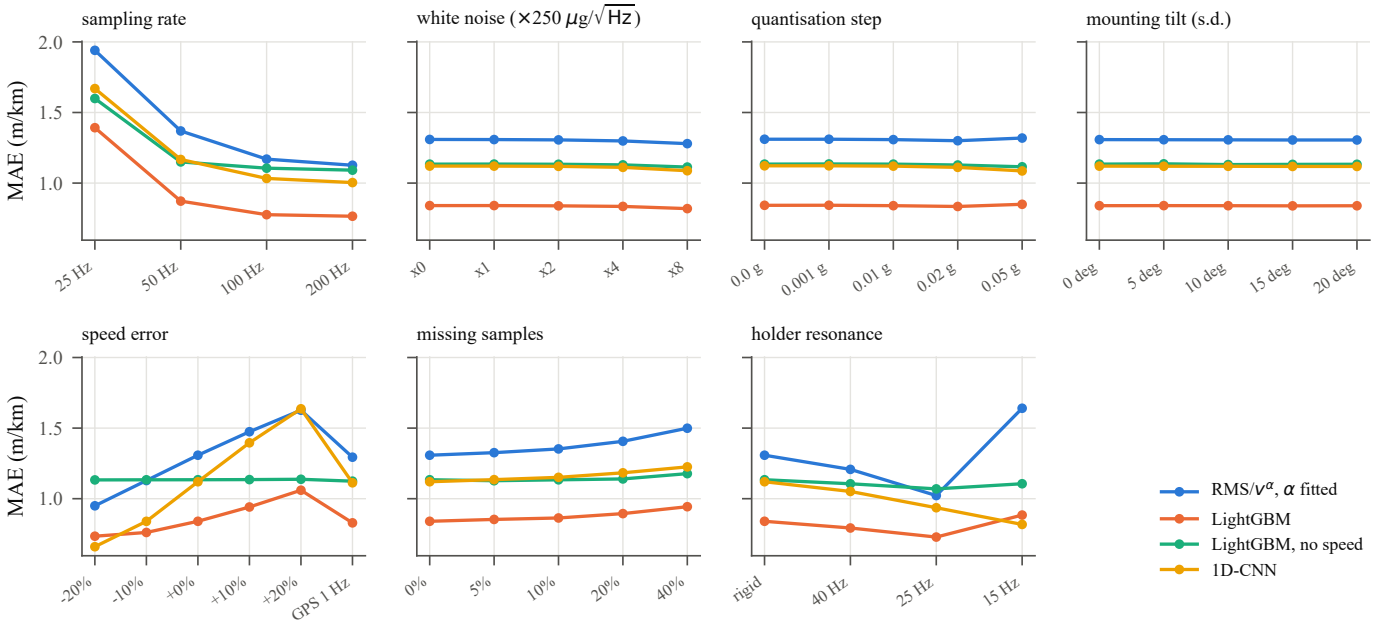


Fig. S7. Sensor robustness in simulation: estimators trained on synthetic profiles, evaluated on 12 held-out synthetic routes driven by 6 unseen vehicles with the telematics sensor configuration; one property changed at a time (common random numbers).

TABLE S27

CONTROLLED SIMULATION (CLASS S; NO REAL DATA). LEFT: NUMBER OF TRAINING VEHICLES AT A FIXED 1,200 TRAINING PASSES (HELD-OUT SYNTHETIC ROUTES, 30 UNSEEN AND UP TO 30 SEEN VEHICLES; LIGHTGBM). Δ: UNSEEN MINUS SEEN MAE RELATIVE TO SEEN, WITH THE UPPER 95 % BOUND FROM A SEGMENT AND FROM A ROUTE BOOTSTRAP; CRITERION OF OUR EARLIER STUDY: UPPER BOUND $\leq$10 %. RIGHT: ONE SENSOR PROPERTY CHANGED AT TEST TIME (LIGHTGBM TRAINED ON SYNTHETIC PROFILES WITH THE SENSOR MIXTURE, TESTED ON 12 HELD-OUT SYNTHETIC ROUTES DRIVEN BY 6 UNSEEN VEHICLES WITH THE TELEMATICS-LIKE SENSOR; MAE AND BIAS IN M/KM).

Vehicles N	Seen MAE	Unseen MAE	Δ (%)	Upper bound (%) segments / routes	Criterion met segments / routes
3	0.62	1.06	+70	76 / 87	no / no
10	0.75	0.92	+24	27 / 31	no / no
30	0.78	0.90	+16	18 / 21	no / no
100	0.82	0.87	+6	8 / 11	yes / no
300	0.81	0.87	+7	9 / 12	yes / no

Sensor property at test time	MAE	Bias
telematics-like, rigid (reference)	0.84	−0.65
+ holder resonance 40 Hz	0.79	−0.55
+ holder resonance 25 Hz	0.73	−0.36
+ holder resonance 15 Hz	0.88	+0.37
sampling rate 200 Hz	0.77	−0.48
sampling rate 50 Hz	0.87	−0.71
sampling rate 25 Hz	1.39	−1.36
speed error +20 %	1.06	−0.98
speed error −20 %	0.73	−0.15
white noise ×8	0.82	−0.58
mounting tilt s.d. 20°	0.84	−0.64
40 % samples lost	0.94	−0.81

TABLE S28
CONTROLLED GENERALISATION TESTS OF THE FIRST ANALYSIS ROUND IN THE PARAMETRIC SIMULATOR, WITH CRITERIA FIXED IN ADVANCE (UNSEEN VEHICLES: RELATIVE MAE INCREASE $\leq$10% INCL. CI; SPEED: |CHANGE| $\leq$10% INCL. CI).

test	model	result	criterion_met
unseen vehicles (random split)	M2a RMS only (alpha=0)	MAE seen 1.365 -> unseen 1.435; diff +0.070 [+0.022, +0.120]	True
unseen vehicles (random split)	M2b RMS/v^0.5	MAE seen 1.066 -> unseen 1.201; diff +0.135 [+0.097, +0.177]	False
unseen vehicles (random split)	M2c RMS/v^alpha (alpha fitted)	MAE seen 1.042 -> unseen 1.200; diff +0.158 [+0.117, +0.204]	False
unseen vehicles (random split)	M2d legacy exponent 2	MAE seen 2.139 -> unseen 2.097; diff -0.042 [-0.110, +0.025]	True
unseen vehicles (random split)	M3 LightGBM	MAE seen 0.742 -> unseen 0.884; diff +0.142 [+0.113, +0.176]	False
unseen vehicles (random split)	M3 LightGBM (no speed)	MAE seen 1.025 -> unseen 1.036; diff +0.011 [-0.026, +0.049]	True
unseen vehicles (random split)	M5 1D-CNN	MAE seen 0.809 -> unseen 1.004; diff +0.195 [+0.167, +0.224]	False
unseen vehicles (random split)	M5 1D-CNN (no speed)	MAE seen 1.137 -> unseen 1.358; diff +0.222 [+0.188, +0.256]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2a RMS only (alpha=0)	MAE low-f_s 1.669 vs high-f_s 1.423; diff -0.246 [-0.332, -0.161]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2b RMS/v^0.5	MAE low-f_s 1.431 vs high-f_s 1.018; diff -0.413 [-0.513, -0.314]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2c RMS/v^alpha (alpha fitted)	MAE low-f_s 1.407 vs high-f_s 0.971; diff -0.437 [-0.531, -0.345]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2d legacy exponent 2	MAE low-f_s 2.270 vs high-f_s 2.141; diff -0.128 [-0.219, -0.039]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM	MAE low-f_s 0.819 vs high-f_s 1.285; diff +0.466 [+0.416, +0.515]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM (no speed)	MAE low-f_s 1.082 vs high-f_s 1.487; diff +0.405 [+0.349, +0.464]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN	MAE low-f_s 0.782 vs high-f_s 1.261; diff +0.478 [+0.377, +0.589]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN (no speed)	MAE low-f_s 1.058 vs high-f_s 1.082; diff +0.024 [-0.038, +0.080]	True
BeamNG: vehicle held out (S2 vs S1)	M0 constant (training median)	MAE seen 7.60 -> unseen 7.60; diff +0.00 [-0.00, +0.01]	True
BeamNG: vehicle held out (S2 vs S1)	M1 speed-only (LightGBM on speed features)	MAE seen 7.49 -> unseen 7.61; diff +0.12 [+0.00, +0.23]	True
BeamNG: vehicle held out (S2 vs S1)	M2a RMS only (alpha=0)	MAE seen 6.68 -> unseen 7.27; diff +0.59 [+0.43, +0.73]	False
BeamNG: vehicle held out (S2 vs S1)	M2b RMS/v^0.5	MAE seen 6.36 -> unseen 6.88; diff +0.52 [+0.37, +0.66]	False
BeamNG: vehicle held out (S2 vs S1)	M2c RMS/v^alpha (alpha fitted on training data)	MAE seen 6.34 -> unseen 6.88; diff +0.54 [+0.40, +0.69]	False
BeamNG: vehicle held out (S2 vs S1)	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	MAE seen 6.50 -> unseen 6.70; diff +0.20 [+0.11, +0.29]	True
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (all features)	MAE seen 6.02 -> unseen 7.25; diff +1.24 [+0.96, +1.51]	False
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (no speed features)	MAE seen 6.01 -> unseen 7.39; diff +1.37 [+1.10, +1.64]	False
BeamNG: vehicle held out (S2 vs S1)	M5 1D-CNN	MAE seen 8.63 -> unseen 10.95; diff +2.32 [+1.78, +2.88]	False
speed 40->100 km/h (sweep_iso)	M2a RMS only (alpha=0)	rel. change +63.3% [+60.8, +65.5]	False
speed 40->100 km/h (sweep_iso)	M2b RMS/v^0.5	rel. change +13.7% [+11.8, +15.4]	False
speed 40->100 km/h (sweep_iso)	M2c RMS/v^alpha (alpha fitted)	rel. change +3.1% [+1.4, +4.7]	True
speed 40->100 km/h (sweep_iso)	M2d legacy exponent 2	rel. change -52.0% [-52.5, -51.5]	False
speed 40->100 km/h (sweep_iso)	M3 LightGBM	rel. change +0.6% [-1.1, +2.5]	True
speed 40->100 km/h (sweep_iso)	M3 LightGBM (no speed)	rel. change +35.7% [+31.9, +38.4]	False
speed 40->100 km/h (sweep_iso)	M5 1D-CNN	rel. change -4.4% [-5.9, -2.5]	True
speed 40->100 km/h (sweep_iso)	M5 1D-CNN (no speed)	rel. change +22.1% [+19.9, +24.2]	False
speed 40->100 km/h (sweep_iso)	raw RMS of a_z (no model, not an IRI)	rel. change +82.5% [+79.1, +85.4]	False
speed 40->100 km/h (sweep_p79)	M2a RMS only (alpha=0)	rel. change +166.7% [+158.7, +171.4]	False
speed 40->100 km/h (sweep_p79)	M2b RMS/v^0.5	rel. change +96.1% [+89.7, +99.9]	False
speed 40->100 km/h (sweep_p79)	M2c RMS/v^alpha (alpha fitted)	rel. change +78.3% [+72.5, +81.8]	False
speed 40->100 km/h (sweep_p79)	M2d legacy exponent 2	rel. change -32.0% [-33.5, -31.1]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM	rel. change +60.2% [+50.4, +67.7]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM (no speed)	rel. change +92.2% [+62.2, +117.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN	rel. change +25.2% [+22.5, +27.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN (no speed)	rel. change +73.1% [+47.9, +94.2]	False
speed 40->100 km/h (sweep_p79)	raw RMS of a_z (no model, not an IRI)	rel. change +235.0% [+223.2, +242.3]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M2c RMS/v^alpha (alpha fitted)	rel. change +7.4% [+5.2, +10.0]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM	rel. change -0.9% [-4.0, +2.0]	True
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM (no speed)	rel. change +35.6% [+16.7, +51.4]	False

S8. Reproduction

Every number in the paper and this supplement is a LaTeX macro generated from the result files (`make_numbers.py`, `make_numbers_v2.py`, `make_numbers_v3.py`); tables and figures are generated by `make_tables.py`, `make_tables_v2.py`, `make_tables_v3.py`, `make_figures.py`, `make_figures_v2.py` and `make_figures_v3.py`. The complete pipeline, including the second and third analysis rounds, is `run_all_experiments.py`; the third round alone (with the M7 fix) is

```
python run_all_experiments.py --force --only v2_physics_inversion v2_summarise \
    v2_numbers v2_tables v3_pooled v3_pooled_merge v3_calib_uncertainty \
    v3_band_limited v3_level_and_bands v3_rank_paired v3_speed_robustness \
    v3_nosim_index v3_pass_density v3_numbers v3_tables v3_figures bib_cited paper
```

`README.md` lists the commands, the pinned environment (`environment_lock.txt`), seeds, runtimes and the provenance of every headline number.